

R&D Employment Spillovers in Trade and MNE Networks: Evidence from a R&D Tax Reform ^{*}

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Abstract

We combine R&D survey data, Swedish and Danish register data, MNE group information, firm-to-firm trade linkages, and a Swedish R&D tax reform that increases subsidies to firms' employment related R&D expenditures to provide evidence for the allocation of R&D activities among Swedish-Danish trade and MNE partners. We first show that Swedish firms increased their R&D employment in response to the introduction of a R&D related tax credit. We then show that the reform leads to an increase in R&D employment among Danish trade partners, but a fall in R&D employment in Danish MNE partners. We conclude that R&D subsidies generate positive international R&D spillovers within trade networks, but relocation of R&D activities across affiliates within multinational companies. Since MNEs account for the majority of total R&D activity, this considerably dampens the reform's effect on global R&D activity.

JEL classifications: F66, F16, J24

Keywords: R&D reform, International R&D spillovers, Firm-to-firm trade and MNE networks.

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1 Introduction

Most countries provide tax incentives for research and development (R&D) activities in private firms, and subsidy rates have increased markedly over the past 25 years (OECD, 2025). The justification for R&D subsidies typically rests on the view that firms tend to underinvest in R&D due to positive externalities not accounted for by the investor. For example, knowledge generated by R&D investment may spill over to and benefit other firms, such that R&D is akin to public good. A substantial body of research highlights the potential benefits of knowledge spillovers due to international trade and Multinational Enterprises (MNEs) for productivity and economic growth (Akcigit and Melitz, 2022; Keller, 2022). On the other hand, R&D subsidies may also be too high if they, e.g., lead to a relocation of R&D activities towards regions or countries with more generous subsidies thereby dampening overall R&D investments (Bloom et al., 2019). However, little empirical evidence exists for R&D activity spillovers within international trade and MNE networks, especially in response to well-defined R&D policy changes that reduce the marginal cost of doing R&D in one country. In this paper, we combine R&D survey data, Swedish register data, Danish register data, MNE group information, firm-to-firm trade linkages, and a plausibly exogenous employment-related R&D policy shock to provide evidence for the allocation of R&D activities among Swedish-Danish trade and MNE partners.

Our empirical approach proceeds in several steps. First, we match Swedish and Danish firms in our register data with detailed MNE membership information. In addition, we employ unique and novel data on firm-to-firm trade transactions between Danish and Swedish firms. Jointly, these data sources identify existing Danish-Swedish trade and MNE network members in combination with their members' register data. This allows us to examine R&D activity based on employment related R&D expenditures, which is the main target of recent Swedish R&D policies.

Second, we take advantage of a new Swedish R&D tax reform as a source of exogenous variation to examine Swedish firm's innovation activities. The reform, introduced in 2014,

allows Swedish firms to deduct employment related R&D expenses but only up to a limit. Consequently, we apply a difference-in-difference identification strategy similar to Bøler et al. (2015). In particular, we compare employment related R&D expenditures across Swedish firms whose marginal R&D expenses are expected to decrease as a consequence of the reform (firms below the cap) to firms whose marginal costs are not affected (firms above the cap) before and after 2014.

Third, we extend the difference-in-difference approach to Danish firms. In particular, focusing on Danish firms with trade and MNE linkages in Sweden, we ask if Danish firms linked to Swedish firms treated by the reform (i.e., Swedish firms below the cap) see a change in their own R&D activity compared to Danish firms who are linked to untreated Swedish firms (i.e., Swedish firms above the cap). In doing so we pay particular attention to whether the type of link, trade or MNE, matters for any spillover effects. While this approach shows whether Danish firms are affected by the Swedish policy through trade and MNE linkages, it does not provide actual substitution elasticities between R&D workers in Swedish and Danish firms. We address this in our fourth and final approach.

Fourth, following an instrumental variable approach similar to Kovak et al. (2021), we estimate the elasticity of Danish firms' R&D activity with respect to their trade and MNE partners R&D activity in Sweden.¹ We start by predicting Swedish firms' R&D activity based on the R&D reform. Then, we use predicted R&D activity in Sweden as an instrument to examine how Danish firms' employment related R&D expenditures respond to the total R&D expenditures by their Swedish MNE and trade partners.

Our results confirm that the Swedish R&D tax reform raises employment related R&D expenditures in Swedish firms below the deduction cap. Furthermore, our spillover result show that Danish firms linked to Swedish trade partners affected by the reform increase their R&D expenditures, while Danish firms linked to Swedish MNE partners that are affected by the reform reduce their own R&D expenditures. In other words, within firm-to-firm trade networks R&D activity is complementary, while it is substitutable across affiliates within

¹Kovak et al. (2021) focus on the effects of offshoring on MNE's employment outcomes.

MNE networks.

These results may reflect that when firms receive employment related R&D subsidies, R&D activities increase and this spills over to trade linked partners, that also raise activities, perhaps due to knowledge spillovers or increased demand in the trade network. However, we also show that the R&D intensity as measured by the wage bill share of college workers rises in Danish firms, so the effect does not only reflect a neutral demand shift. The negative spillover effect in MNE linked firms, on the other hand, reflect at least partly that MNEs relocate tasks within the corporation such that R&D activities are shifted to the location with lowest marginal costs. The data allow us to track workers moving across the border, and we show that treated Swedish MNEs are more likely to hire workers previously employed by affiliates in Denmark than untreated Swedish MNEs.

Our elasticity estimates provide evidence for magnitudes. The estimates combined with back-of-envelope calculations predict that the Swedish tax reform led to the hiring of ten additional R&D workers in Swedish firms independent of the firms' MNE or trade relationships. However, in response to the same increase in R&D activity in Sweden, Danish MNE partners lay off four employees with college degrees, while Danish trade partners hire an additional 0.2 workers. Therefore, our results provide evidence that MNE and trade networks transmit common R&D shocks in one location quite differently to other locations depending on the nature of the link. In particular, trade networks increase their overall R&D activity in response to tax incentives in one location, while MNE members are more likely to substitute R&D activities across locations.

Our results contribute to several strands of the literature. First, our work is related to the literature evaluating the impact of R&D tax reforms. Early studies have used aggregate data to examine the impact of tax reforms on R&D activity, see Becker (2015) for a survey. A part of the literature has evaluated if R&D subsidies attract investments at the expense of other locations and has found some evidence in support for this mechanism.² Recent work

²In their survey of the literature, Bloom et al. (2019) conclude that "...despite some relocation across place, the aggregate effect of R&D tax credits at the national level both on the volume of R&D and on productivity is substantial." Examples from the literature examining relocation effects include Akcigit et al. (2016), Akcigit et al. (2022), Bloom and Griffith (2001), Lucking (2019), Moretti and Wilson (2017) and

use firm-level data and quasi experimental tax variation to show that R&D activity rises in firms treated by the reforms, and we mention the two most closely related here.³ Bøler et al. (2015) examines the impact of a 2002 Norwegian R&D tax reform on R&D expenditures, imported inputs and firm performance using a Difference-in-Differences model and find positive impacts. The Norwegian reform is similar to the Swedish reform we study in that it provided firms a R&D tax credit, but only up to a threshold, such that marginal cost of R&D were lowered only among firms below the cap.⁴ Dechezleprêtre et al. (2023) examines the impact of a UK tax reform and find evidence for positive effects on patenting and are the first to document positive spillovers (to technologically close domestic firms) using a R&D tax reform for identification.⁵ Compared to this literature we confirm the positive effects on R&D activity in treated local firms, but we examine *international* R&D employment spillovers in firm networks and show that the sign of the spillover effects depends on the nature of the firm-to-firm link (trade or MNE). We highlight a new and quantitatively important channel to this literature (relocation across countries within multinationals) through which R&D spillovers are negative.

Second, there is a substantial literature linking knowledge spillovers to international trade, see, e.g., Akcigit and Melitz (2022) and Keller (2022) for recent reviews. Early contributions provided country and industry level evidence on R&D spillovers, see, e.g., Coe and Helpman (1995) and Keller (2002). More recent works have used firm-level data, more credible identification approaches and zoomed in on more specific channels for knowledge spillovers. For example, Atkin et al. (2017) conduct a randomized controlled trial to document "learning-by-exporting" effects among Egyptian rug producers. Aghion et al. (2023) employ an event study approach and use French firm-level export and patent data to show that destination countries are more likely to cite a firm's patents after entry into the destination

Wilson (2009).

³Other papers in the literature using micro level data and quasi-experimental variation in tax incentives include Agrawal et al. (2020), Akcigit et al. (2022), Bajgar and Srholec (2025), Chen et al. (2021), Guceri and Liu (2019) and Rao (2016).

⁴Unlike the Norwegian reform, the Swedish reform is targeted towards R&D *employment* expenditures.

⁵Earlier papers studying R&D spillovers include, e.g., Bloom et al. (2013), Griliches (1992), and Jaffe et al. (1993). Andrieu and Morrow (2025) is a more recent study that evaluates R&D subsidies in France and finds positive up- and downstream spillovers using input-output tables.

country. Another related branch of the trade literature examines transmission in firm-to-firm networks, see Bernard and Moxnes (2018) for an early review.⁶ Most of this literature focus on trade frictions, pricing strategies or transmission of demand shocks in the trade network. Relative to the trade-and-spillover literature, our contribution is to show how a R&D tax reform generates R&D employment spillovers through firm-to-firm trade networks.

Third, we also contribute to an emerging literature examining R&D activity in MNEs. Bilir and Morales (2020) use structural estimation to show that R&D activity among U.S. parent firms and their affiliates is complementary. Gumpert et al. (2025) also provide evidence for complementarities within MNEs in the sense that affiliates are found to file a large share of MNEs’ patents. Along the same lines, Bircan et al. (2024) show that patents are increasingly created in cross-border collaborative teams of inventors within MNEs. Knoll et al. (2021) use European MNE data to provide (non-causal) evidence that MNE’s patent citations follow local tax incentives but conclude that R&D activity is substitutable across locations.⁷ We provide several contributions to this literature. First, in contrast to these papers, we develop a plausibly exogenous identification strategy rooted in the details of a recent policy initiative. This mitigates concerns that current or future R&D policy may respond to existing domestic or foreign R&D activity. This approach is important because in line with several of the papers above, we find a positive correlation between R&D activities across affiliates within MNEs in the raw data, but once we exploit the exogenous variation provided by the R&D reform, it turns negative. Second, we jointly examine trade and investment related mechanisms.⁸ This is relevant from an identification point of view, because knowledge is embodied in trade flows and trade relationships may be correlated with MNE activities.

Fourth, a vast literature investigates if foreign direct investments (FDI) generate spillovers

⁶More recent works include Demir et al. (2024), Dhyne et al. (2021), Dhyne et al. (2025), Eaton et al. (2025) and Heise (2024).

⁷A related literature, which is not focused on R&D employment, examines how changes in labor market policies or corporate tax rates affect employment across affiliates within MNEs, see, e.g., Ahn et al. (2025) and Kovak et al. (2021).

⁸The literature tends to exclusively focus on MNE networks, FDI, or trade. However, recent research highlights the importance of jointly examining various globalization mechanisms in the context of patent related innovation activities (Aghion et al., 2024). A notable exception is Keller and Yeaple (2009).

to local firms. The evidence points to a mix of negative domestic spillover effects (e.g., Aitken and Harrison, 1999; Hu et al., 2005) and more recently positive effects (e.g., Eppinger and Ma, 2024; Haskel et al., 2007; Keller and Yeaple, 2009; Malikov and Zhao, 2023; Setzler and Tintelnot, 2021). Some papers find that labor mobility is important for MNEs to generate local spillover effects, i.e., former MNE employees bringing new knowledge to local firms to improve productivity (Babina and Howell, 2024; Balsvik, 2011; Poole, 2013). Consistent with this mechanism, our evidence focuses on a policy that reduces employment related R&D expenditures that affects firms hiring decisions. Based on this identification strategy, we find that employee related R&D activity is relatively footloose within MNEs. Consequently, local spillover effects based on employee turnover likely depend on where MNEs decide to engage in R&D activity.

Finally, our paper is also related to a line of research documenting that MNEs tend to exploit corporate tax rate differentials through profit shifting and other initiatives, see, e.g., Dharmapala (2019), Huizinga and Laeven (2008), Laffitte and Toubal (2022) and Tørsløv et al. (2022). We complement this literature by showing that MNEs relocate R&D activities in response to changing marginal costs of R&D. MNEs not only move profits to tax havens and other countries with low corporate tax rates. They are also in position to shift R&D activities to countries offering the most generous R&D subsidies. This is important in light of the fact that MNEs account for the majority of R&D activities worldwide (Bilir and Morales, 2020) – in our data MNEs account for roughly two-thirds of total R&D expenses. This suggests that R&D subsidies may be too high seen from a global perspective if R&D policies mainly are designed to attract R&D activity at the expense of other countries.

The rest of this paper proceeds as follows. Section 2 provides background on the Swedish R&D policy reform. Section 3 discusses data sources and data handling procedures. We discuss the empirical specifications and identification strategies in Section 4 before we show how the reform affects Swedish firm in Section 5. Section 6 presents spillover effects on Danish firms, and finally Section 7 concludes.

2 R&D Tax Reform in Sweden

In January 2014 Sweden implemented an R&D-related tax reform that allows firms to deduct R&D-related employment expenses. The goal of the reform is to subsidize the cost of employees working in R&D to increase R&D activity and overall growth (*SOU 2012:66* n.d.). The policy was partly a reaction to similar earlier legislation in other countries, e.g. SkatteFUNN in Norway, that was examined by Bøler et al. (2015). This section provides background information related to the reform and discusses how the reform may affect firms' marginal R&D expenses.

The R&D tax deduction allows firms to deduct 10% of wages against the payable employee contributions. For example, for an employee with a monthly salary of 50,000 SEK, firms can reduce their monthly employer contributions by just under a third (from 15,710 to 10,710 SEK) and the benefit to the firm is immediate, i.e. in the month the firm claims the deduction. The deduction is designed to be simple to administer, and it is estimated that the administrative costs per person-year are only 1,000 SEK (*SOU 2012:66* n.d.).

Firms in all sectors and size categories are eligible for the R&D deduction. However, to qualify for the deduction, the employee must work 75% of their time with R&D and be between the ages of 26 and 65. Furthermore, the research must be systematic, and for commercial purposes to innovate new products or to substantially improve old products.

The tax reform stipulates a maximum deduction of 230,000 SEK per month or 2.76 million SEK per year. Consequently, if a firm has high employment related R&D expenditures sufficient to qualify for the maximum deduction, then claiming the deduction does not affect the cost of hiring additional workers. On the other hand, if a firm has relatively low employment related R&D expenditures such that it does not initially qualify for the maximum deduction, then claiming the deduction reduces the cost of hiring additional R&D workers. Therefore, following Bøler et al. (2015), we expect that the R&D tax deduction reduces a firm's marginal cost of R&D related employment expenses only if the firm's existing employment related R&D expenditure is sufficiently low such that it does not qualify for

the maximum deduction.⁹

Table 1 provides summary statistics highlighting the Swedish tax reform's potential to reduce firms' marginal employment expenses based on Swedish R&D survey data (see Section 3.1 for details). For each survey year the table reports the number of reporting firms in column 1, and the share of firms reporting positive R&D employment expenditures in column 2. Then, column 3 reports the share of firms with sufficiently low R&D employment expenditures such that they would not qualify for the maximum deduction. Recall, a Swedish firm j does not qualify for the maximum deduction, if 10% of its employee related R&D expenditure in year t , $R\&D_{jt} \times 10\%$, falls below the deduction limit of 2.76 million SEK, i.e, if $R\&D_{jt} < 27.6$ million SEK. Of course, firms cannot take advantage of the deduction in the pre-reform periods 2011 and 2013. However, based on observed R&D expenditures in those years we can compute the share of firms with sufficiently low R&D employment expenditure such that they would not qualify for the maximum deduction, i.e., the share of firms with R&D expenditures that meet the restriction $R\&D_{jt} < 27.6$ million SEK. In the pre-reform period, we interpret this number as the share of firms for which we expect to see a reduction in marginal employment expenses due to the R&D tax reform based on their sufficiently low R&D expenditure in 2011 and 2013.

⁹Note that the deduction applies to firm groups.

Table 1: R&D employment expenditures in Sweden according to R&D survey

	(1)	(2)	(3)	(4)	(5)	(6)
	#Firms	Firms with R&D > 0 (%)	Firms with R&D < 27.6 M (%)	Firms with R&D < 27.6 M, cond. on R&D > 0 (%)	Mean R&D cond. on R&D > 0 (M SEK)	Median R&D cond. on R&D > 0 (M SEK)
2011	6,951	13.3	97.7	83.1	44	5.0
2013	7,232	14.6	97.6	83.6	41	4.4
2015	6,789	18.0	97.4	85.8	38	3.8
2017	6,733	15.7	97.4	83.6	50	4.4
2019	7,781	13.9	97.6	83.5	51	4.9

Notes: The table presents data from the Swedish biennial R&D survey for the years 2011, 2013, 2015, 2017, and 2019.

Column 1 shows that the number of firms that report in the R&D survey is relatively stable across the survey years. Column 2 shows that a substantial number of firms report zero R&D employment expenses. A potential reason for this is that R&D involves fixed costs such as equipment, space, and laboratories, as well as a fixed investment to develop the know-how to effectively engage in R&D and take new innovations to the market.

Since most firms report zero R&D employment expenses, it follows that most firms report R&D employment expenditures below the cap in the pre-reform years (column 3), but even among firms that report positive R&D employment expenditure pre-reform, around 83% are below the cap (column 4). Columns 5 and 6 show that the mean R&D employment expenditures among firms reporting positive expenditures are much higher than the median reflecting that a few firms have very large expenditures.

2.1 R&D Policy Changes in Denmark and Other Countries

Sweden is not alone in subsidizing R&D expenditures. 33 out of 38 OECD jurisdictions offered tax relief on R&D expenditures in 2024, and the average OECD marginal R&D tax subsidy rate more than tripled from 2000 to 2024 (OECD, 2018). The Swedish R&D tax policy places the country in the middle among OECD countries, while Denmark's R&D

policies are somewhat less generous.

Denmark implemented a distinct R&D tax reform in January 2012, the so-called “Skatte kreditordning”, in which tax credits are given to loss-making private R&D firms for the part of losses that can be attributed to R&D expenses. The aim of the reform was to strengthen the liquidity of small firms in their start-up phase when R&D investments have not yet yielded a return. Given the focus on small loss-making firms, the different timing of the reforms and the rather small scale of the Danish reform in comparison with the Swedish reform, we do not believe there are important confounding effects on any spillovers from the Swedish R&D reform.

In addition, the Danish government also decided to provide additional incentives for R&D by gradually increasing the standard 100 percent immediate write off of R&D expenses (as is common in most countries) to 110 percent from 2019 to 2026. Many countries offer such superdeductions for R&D expenses (Bloom et al., 2019). Again, given the staggered timing and the gradual introduction of the additional R&D tax credit, we do not believe this impacts on estimated spillovers to Danish firms from the Swedish reform.

3 Data

This section describes our data sources and explains how we relate various samples to the Swedish R&D policy reform. All samples are available for the period 2008-2019.¹⁰ In addition, we summarize our procedures for identifying trade and MNE links between Danish and Swedish firms to examine spillover effects.

3.1 R&D Activity by Swedish Firms

We observe Swedish firms’ R&D activity based on a biennial survey from 2011 to 2019 (Sv. Forskning och utveckling). The survey is collected in line with the Frascati manual and covers around 95% of the officially reported R&D expenditures in the Swedish business

¹⁰The Swedish 2009 R&D survey did not include a question about R&D employment expenditure and is therefore not included in Table 1.

sector.¹¹ The data includes firm identifiers that can be merged with other data sources.

The R&D survey includes: (i) all firms with at least 200 employees, (ii) all firms that have 10 or more employees within the sector Scientific research and development (SNI sector code 72), (iii) a sample of firms with 10-199 employees that are randomly selected conditional on size and sector. Firms that had over 5 million SEK in the previous R&D survey are always included in the survey. This sampling strategy leads to attrition among the smallest firms in the sample. Overall, the survey response rate is high (around 92% in both 2011 and 2013).

The survey collects data on both intra- and extramural R&D expenditures. For our purposes, we focus on the largest component of R&D expenditures: intramural labor costs of R&D personnel.¹² Throughout, we simply refer to this component as "R&D employment expenditures".

In summary, the Swedish R&D survey provides useful information to examine the effect of the R&D reform on R&D activity, i.e., based on R&D expenditures in the pre-reform period we can examine if firms that are most likely to see a decrease in marginal employment costs due to the tax deduction will indeed increase R&D activities. However, it is not clear based on the employment expenditures in the R&D survey how the reform affects employment opportunities and how to examine if potential effects translate to Danish partner firms. To address these challenges we work with firm-level employment data which we will introduce in the next subsection.

3.2 Swedish Linked Employer-Employee Data

One potential challenge measuring R&D activity from surveys is that firms may find ways to relabel expenses in order to inflate R&D expenditures. For example, Chen et al. (2021) estimate the impact of a change to the Chinese tax scheme and find positive impacts on

¹¹This can be seen from aggregating the expenditures on own R&D (variable v207) and comparing to the official aggregate data posted by Statistics Sweden, see series called Total intramural expenditure on R&D.

¹²According to the Frascati manual, these labor costs should include salaries, associated costs or fringe benefits, travel costs, payroll taxes and social security payments. This applies to both personnel that works full and part-time with R&D.

R&D expenses, but a sizable share of the reported R&D was found to be relabeling. We can get around such issues by using matched worker-firm data and detailed education codes from administrative registers of the workers. Firms cannot manipulate education codes of their workers. In addition, the register data also allows us to extend the sample window to 2008-2019 including years where the R&D surveys are not conducted, thereby greatly increasing the sample size.

We use the Swedish register LISA which includes linked employer-employee data for the total population of Swedish firms including workers' education for the sample period, 2008-2019. We use this information to compute firms' total wage bill as well as annual salary payments to educational subgroups of workers including those with college degrees, STEM degrees and PhD degrees. In our main specification we will use firm-level salaries paid to college workers as outcome variable. We label this variable $\text{College-Salary}_{jt}$.¹³

Our identification strategy requires that we observe the firms in the population that have experienced a reduction in the marginal cost of hiring additional R&D professionals as a consequence of the 2014 R&D reform. Therefore, we merge the population of Swedish firm-years with the $\text{R\&D}_{jt}$ variable constructed in the R&D survey data (see Section 2). The merge between the R&D survey and the linked employer-employee data is nearly perfect, as both data sources have the same unique firm identifiers. However, firms' inclusion in the R&D survey depends on the survey's sampling strategy. As a consequence, we can map firms' R&D expenditures from the R&D survey to the register data only for a subset of firms. In particular, for our estimation sample, we keep only those observations of firms for which we observe at least some information about R&D employment expenditures prior to the reform.

Table 2 provides summary statistics for college worker salaries for the population (excl. public administration) of firms with at least one college worker and for the estimation sample, where we also observe the level of pre-reform R&D expenditures. The R&D survey targets firms that plausibly engage in R&D activity. Therefore, conditioning on observing R&D

¹³In extensions we also use the number of college workers, STEM workers and workers with PhD degrees as outcome.

expenditure drops a large fraction of the firm population. However, as we would expect, the firms we drop are small. The resulting estimation sample consists of about 8,000 firms per year with an average total wage bill paid to workers with a college degree of about 20 million SEK.¹⁴ In the average year, the sample constitutes more than 30 percent of college worker salaries in the population. Restricting attention to firms with at least one college worker maintains more than 99 percent of population-wide R&D expenditures.

Table 2: Swedish firms with college employees, population and estimation sample

	Population			Estimation sample		
	#Firms	Mean college salaries (M SEK)	Total college salaries (B SEK)	#Firms	Mean college salaries (M SEK)	Coverage of total college salaries (%)
2010	141,966	3.5	493	8,598	18	31
2011	147,680	3.6	525	8,810	19	32
2012	152,554	3.6	553	8,790	20	32
2013	157,762	3.7	578	8,670	21	32
2014	163,309	3.7	608	8,358	23	31
2015	168,100	3.8	642	8,058	24	31
2016	172,422	4.0	682	7,813	26	30

Notes: "Population" is all Swedish firms (excl. public administration) with at least one employee with a college degree. "Estimation sample" is the subset of firms that appear in the biennial R&D survey prior to the reform (that is in 2011 and/or 2013).

Based on the effects of the R&D reform on college workers' salaries in Sweden, we examine spillover effects to the Danish labor market by examining Danish firms' expenses on college workers. To do so, we establish trade and MNE linkages between Swedish and Danish firms in the following two subsections.

¹⁴The yearly number of firms in the estimation sample (Table 2) exceeds the yearly number of firms in the R&D survey (Table 1). This is because we only require firms in the estimation sample to be present in the R&D survey in some pre-reform year, that is, 2011 or 2013.

3.3 Trade Links

We use data on firm-to-firm export flows within the EU collected by the tax authorities, Skatteverket in Sweden and Skat in Denmark, to determine if a Swedish-Danish firm pair trades with each other in any given year (2012-2016). Firms that trade with each other in the EU must declare the national firm identification number of their foreign trading partner as well as the value, and type of trade (trade in goods, services, or third-party trade). In practice, the exporter can sell a good or a service without VAT within the EU if they provide a valid VAT number of the foreign buyer on the invoice. The exporter then reports these sales to the local tax authorities (Skat/Skatteverket). Buyers (importers) are then responsible for calculating and paying the VAT in the destination country. By combining this data for both Denmark and Sweden we are able to construct cross-border firm-to-firm trade linkages between firms in Denmark and Sweden. The quality of these data is nearly complete due to the responsibility of the seller to provide a valid VAT number on the invoice. These VAT numbers can for example be validated through an EU VIES search service, which checks the validity of the firm identifier in the foreign country.

3.4 MNE Links

We observe if any given Swedish-Danish firm pair is a member of the same MNE network. If both firms in a given pair are members of the same multinational network, then we consider these firms linked by MNE ownership.

We use two main datasets to construct these MNE linkages. First, the Bisnode Group Register (now part of Dun & Bradstreet) reports all domestic and international members of MNE groups that contain a Swedish firm. Second, Experian provides comparable information for Danish firms. We combine Bisnode and Experian to determine which Danish and Swedish firms are part of the same MNE firm group.

In particular, each MNE firm group network consists of one group mother and subsidiaries that are owned directly or indirectly by the group mother. The ownership structure is

organized through chains of mother-daughter firm relations.¹⁵ For example, if firm A owns firm B, and firm B owns firm C, then we consider all three firms (A, B, C) members of the same MNE network. Note that we consider domestic and foreign MNE groups. For example, a foreign firm A may own Danish firm B and Swedish firm C. We then consider B and C members of the same MNE network independent of whether A is a Swedish firm, a Danish firm, or a firm located in a third country. Therefore, we take advantage of direct and indirect as well as domestic and international ownership structures to identify MNE groups and MNE ownership links of Danish and Swedish firms.

All Swedish firms in Bisnode and Danish firms Experian are identified by a domestic ID number that can be linked to Swedish and Danish register data. Foreign firms have either a name and a time-consistent ID (not an official local identification number). To enable us to merge these data, we perform a matching of Bisnode/Experian with local registers to add the local firm identifier. Overall, the match rate from the raw data is high both for the Bisnode and Experian data. For Bisnode, we are able to assign 99% of the Danish firms with an official Danish firm identifier (CVR number). Similarly, for Experian, we are able to match around 91% of the Swedish firms with an official Swedish firm identifier (organisationsnummer). In addition, we are able to identify firms that are linked via a third country. In our data, we are able to identify 1,584 firms in third countries that are linked to 2,129 Danish and 5,566 Swedish firms. Appendix B provides additional details regarding the matching process.

To examine the spillover effects of the Swedish R&D reform to Danish R&D activity, we use the trade and MNE links developed in the previous two subsections to Danish employer-employee data explained in the next subsection.

3.5 Danish Linked Employer-Employee Data

As for the Swedish firms we have access to the population of firms in Denmark and their linked workers. We start with the register of Danish employer-employee-linked data (BFL)

¹⁵Each relation must be at least a 50% ownership stake.

which we link to UDDA, a register containing information on the educational level of residents. For each Danish firm, i we then generate the total salaries paid to workers with college degrees: $\text{College-Salary}_{it}$.

Next, using the trade links and MNE links established in Sections 3.3 and 3.4, we determine which Danish firms were potentially affected by the Swedish R&D reform because their Swedish trade and MNE partner were affected. We do this in three steps. First, we drop all Danish firms that, according to our trade and MNE links, do not have any trade or MNE partners in the Swedish sample developed in Section 3.2 prior to the Swedish tax reform in 2014. Second, to this sample of Danish firm-years we merge all of their Swedish trade and MNE partners' college salaries and their R&D employment expenditures, $\text{R\&D}_{jt}$. In the resulting sample, for each Danish firm i we now observe all of i 's trade partners, MNE partners, including their partners' expenditure on college salaries and employment related R&D expenditures based on the Swedish R&D survey, $\text{R\&D}_{jt}$.

Table 3 reports summary statistics for the population of Danish firms with college workers and the estimation samples of Danish firms with trade or MNE links. In the population (excl. public administration), the average firm pays just over four million DKK in salaries to its workers with college education. About every fourth firm is connected to a Swedish firm via trade or MNE. Conditional on an existing trade or MNE relationship, the average firm has just below six Swedish partners, whereas the median firm has two (not reported in Table 3).

Table 3: Danish firms with college employees, population and estimation samples

Population			Estimation sample, any relationship				Estimation sample, MNE relationships			
	#Firms	Mean college salaries (M DKK)	#Firms	Mean college salaries (M DKK)	Mean links cond. on some	Median links cond. on some	#Firms	Mean college salaries (M DKK)	Mean links cond. on some	Median links cond. on some
2010	83,611	4.1	11,919	21	-	-	1,007	35	-	-
2011	84,786	4.1	12,530	20	-	-	1,036	36	-	-
2012	86,328	4.2	12,919	21	3.4	1	1,015	94	3.2	1
2013	87,994	4.3	12,842	21	3.4	1	1,004	93	3.1	1
2014	90,354	4.3	12,388	23	3.8	2	940	88	2.9	1
2015	94,133	4.3	12,028	24	3.9	2	910	84	2.8	1
2016	98,363	4.4	11,581	26	4.0	2	889	81	2.7	1

Notes: "Population" consists of all Danish firm-years with college employees. These are then merged with the population of Swedish firm-years with college employees (presented in Table 2). "Estimation sample" restricts Danish firms to the Danish estimation sample and Swedish firms to the Swedish estimation sample (as outlined in Table 2). "Estimation sample, MNE relationships" differs from "Estimation sample" by ignoring trade links and only considering MNE links.

Our estimation sample contains around 12,000 firms each year, corresponding to 13.5% of the population. Firms in the estimation sample pay about five times more to workers with college education, i.e., Danish firms with trade and/or MNE partners in Sweden are relatively large.

Conditional on having some relationships, the mean number of relationships is 3.4-4.0, while the median is 1-2. This is lower than for the population because we now only consider relationships that enter the Swedish estimation sample. Lastly, we have a particular interest in Danish firms with a Swedish MNE relationship prior to the reform. These constitute around 900-1,000 Danish firms each year, with a mean of around three Swedish MNE counterparts, while the median is one.

3.6 Descriptive Statistics

This section provides some relevant descriptive statistics. First, since our estimation samples include all private firms, it is worth pointing out the industries with most firms. Appendix Table A1 shows that most Swedish firms with trade or MNE links to Danish firms are in the Manufacturing sector, while Wholesale & Retail and Professional, Scientific, and Technical Activities also account for sizable portions of the sample. Among the connected Danish firms Manufacturing also is important but Wholesale & Retail accounts for a relatively larger share than among the Swedish firms. It is also evident from Appendix Table A1 that

the vast majority of firms employ at least one college worker, while most firms do not have workers with PhD degrees on their payroll.

Second, Denmark is an important trade partner for Sweden. Appendix Table A2 shows that among Swedish firms that export to the EU, 72 % also export to Denmark and that Denmark accounts for 31 % of the export value while 33 % of Swedish firms have Denmark as the top export destination. The corresponding numbers for Swedish firms that import from Denmark slightly lower.

Third, we ask how much of total Swedish R&D employment expenditures and R&D expenditures of any kind are accounted for by MNEs and non-MNEs that trade (to any country, not just Denmark). In addition, we are also interested in how much of total R&D activity is accounted for by Swedish firms with MNE or trade links to Danish firms (i.e., the estimation sample).

Fourth, as explained in Section 2 we exploit that firms below the cap of SEK 27.6 M had their marginal costs of doing R&D reduced. This may raise concerns about endogenous sorting of firms before and after the introduction of the R&D tax credit. However, Appendix Figure A1 show no clear sign of bunching around the cutoff in any years before or after the reform.

Finally, we examine how employment in Swedish firms correlate with employment in connected Danish firms. Appendix Figures A2 and A3 show positive correlations for both the total wage bill and the college worker wage bill. This is true for both trade and MNE connected firms and irrespective of whether industry fixed effects are included or not. These correlation patterns for our measure of R&D activities, the college worker wage bill, are consistent with how R&D activity covaries across affiliates in Bilir and Morales (2020), Bircan et al. (2024), and Gumpert et al. (2025). However, we are interested in how these patterns may change when we use quasi experimental variation that reduces the marginal costs of doing R&D in one location. We turn to how we identify the effects of the Swedish R&D reform next.

4 Identification

Our goal is to investigate if policy-driven R&D activity by Swedish firms affects R&D activity by their Danish trade and MNE partners. To accomplish this, we will first examine the effect of the R&D tax reform in Sweden on Swedish firms. Next, we study R&D spillovers from Swedish to Danish firms via trade and MNE linkages.

4.1 Swedish R&D Reform, R&D activity, and R&D Employment

Section 2 explains that the tax reform only has the potential to lower marginal costs of employment related R&D activity for Swedish firms with sufficiently low employment related R&D expenditures, such that they do not qualify for the maximum deduction. The Swedish R&D survey does not report actual take up of the tax credit, so we follow an intention-to-treat type identification strategy also adopted by Bøler et al. (2015).

This empirical strategy takes advantage of the fact that, we can identify the firms that are eligible for the tax credit according to pre-reform employment related R&D expenses as measured in the R&D survey. Let $R\&D_{jt}$ be R&D activity measured as R&D related employment costs reported by Swedish firm j in year t in the R&D survey. Recall from Section 2 that the cost of additional employment related R&D expenditures is only affected by the R&D reform if $R\&D_{jt} < 27.61$ million SEK, i.e., if the firm does not qualify for the maximum deduction. We then consider a firm treated by the R&D reform if the firm is observed not to qualify for the maximum deduction based on pre-reform R&D expenditures.

In particular, let $1\{t \geq 2014\} = 1$ denote an indicator variable for the year of the R&D reform and subsequent years. In addition, let $H_j = 1$ if firm j 's average annual R&D expenditure over the pre-reform period is below 27.6 million SEK and 0 otherwise. Combined, if $1\{t \geq 2014\} \times H_j = 1$ then we consider firm j treated by the R&D reform in 2014 and beyond.

We then empirically examine if the R&D reform increased R&D activity by estimating

the following difference-in-differences model:

$$\log \text{R\&D}_{jt} = \mu_j + \eta_t + \gamma 1\{t \geq 2014\} \times H_j + \varepsilon_{jt}. \quad (1)$$

The coefficient of interest, γ , measures the response of R&D activity of firms that are potentially affected by the R&D reform. We consider this an intention-to-treat effect.

The firm fixed effect, μ_j , and the time effect, η_t , account for firm and time-specific omitted variables. The identification assumption is that conditional on these fixed effects, firms' variation in R&D expenditure as captured by the error term ε_{jt} is not systematically related to treatment $1\{t \geq 2014\} \times H_j$, and that the R&D performance of firms affected by the reform follows the same trend compared to firms that are not affected prior to the reform. Without the fixed effects, the model would likely not be identified. For example, the coefficient estimates for γ could then simply capture that smaller firms with less R&D expenditure are more likely to be affected by the reform. Furthermore, the time effects η_t capture overall trends.

To examine if the R&D reform affects firms that trade with Danish firms or firms that are linked by MNE ownership, we estimate equation (1) over the entire sample of the R&D survey as well as a sample that focuses only on firms that are linked to Danish firms by either trade or MNE links in the pre-reform period. To examine if MNE links in particular are affected, we augment equation (1) with an interaction term. Let $MNE_j = 1$ if Swedish firms j is linked to a Danish firm as explained in Section 3.4 and 0 otherwise. We then estimate

$$\log \text{R\&D}_{jt} = \mu_j + \eta_t + \gamma_1 1\{t \geq 2014\} \times H_j + \gamma_2 1\{t \geq 2014\} \times H_j \times MNE_j + \varepsilon_{jt}. \quad (2)$$

In this specification, γ_1 captures differences in R&D performance between potentially treated firms and non-treated firms. The coefficient γ_2 allows for differences for firms that are linked to Danish firms via MNE ownership.

To examine pre-trends and persistence of the reform effect, we also estimate an event

study version of the model. That is, we allow the reform effect to change over the sample window, 2008-2019, and estimate

$$\log \text{R\&D}_{jt} = \mu_j + \eta_t + \sum_{k=2008, k \neq 2013}^{2019} \gamma^k H_j + \varepsilon_{jt}. \quad (3)$$

We apply the same empirical models to examine the effects of the R&D reform on employment of college workers. That is, we replace the outcome variable in equations (1), (2) and (3) with $\log \text{College Salary}_{jt}$, i.e., the salaries paid to workers with college education by firm j in year t . Although the empirical model remains similar to equation (1), the estimation sample will change, as we explained in Section 3.2. We observe college salaries in every year, and they are much less likely to be zero (with non-existing logs) compared to R&D expenditure. On the other hand, to apply our identification strategy, we still need to observe pre-reform R&D expenditures in the Swedish R&D survey.

In this case, the reform effect measures the response of employment of college workers of firms that are affected by the R&D reform. From an interpretation point of view, this captures R&D effects of the reform more broadly than when using R&D employment expenditures from the surveys as outcome. For example, firms that increase R&D employment expenditures in response to the reform, may hire additional workers with college degrees to support R&D activities for example via improved management.

4.2 Spillover Effects

Next, we develop an empirical strategy to examine if potential R&D effects in Sweden spill over to connected Danish firms. We rely on the trade and MNE links we explained in Sections 3.3 and 3.4 combined with a reduced form approach and an instrumental variable identification strategy similar to Kovak et al. (2021).

4.2.1 Reduced Form Approach

Our reduced form approach extends the treatment indicator from the previous sections to Danish firms. In this identification approach, we compare Danish firms that are linked to Swedish firms (trade or MNE) that are affected by the Swedish R&D reform according to the R&D survey information (i.e. $H_j = 1$) to Danish firms that are linked to Swedish firms (trade or MNE) and are not affected by the Swedish R&D reform (i.e. $H_j = 0$). We start by specifying the relevant treatment indicator and then explain the preferred empirical model.

Let Ω_i be the set of Swedish firms that Danish firm i is linked to within the pre-reform period by trade or MNE relationships. We consider a Danish firm i affected by the Swedish R&D reform if its Swedish counterparts are affected by the reform. In particular, recall that $H_j = 1$ if a Swedish firm's R&D expenditure in the pre-reform period is sufficiently low such that its marginal costs of hiring additional R&D employees are affected by the reform. We then let $D_i = 1$ if Danish firm i is linked to at least one firm in the Swedish R&D survey that is (intentionally) treated by the Swedish R&D reform, i.e., $D_i = \max_{\Omega_j} H_j$, and zero otherwise. Finally, we combine the indicator D_i that identifies the treatment status of Danish firms with the pre/post-reform indicator to examine if R&D reform effects in Sweden translate to Danish R&D activity as measured by salaries paid by Danish firms to workers with a college education. That is, we estimate the empirical model:

$$\log \text{College Salary}_{it} = \mu_i + \eta_t + \delta 1\{t \geq 2014\} \times D_i + \varepsilon_{it} \quad (4)$$

The coefficient of interest δ reveals the R&D response of a Danish firm that is linked to a Swedish firm that saw a potential decrease in its marginal cost of R&D activity due to the Swedish R&D reform relative to Danish firms that are linked to Swedish firms that did not experience the same marginal cost reduction.

If $\delta > 0$, then we conclude that R&D activity across linked firms is complementary. If $\delta < 0$, then we conclude that investment into R&D is a substitute across linked firms in Sweden and Denmark. The results section will focus on different samples to examine if R&D

activities across trade and MNE partners are substitutes or complements.

Finally, to check if pre-trends are aligned for Danish firms linked to treated and untreated Swedish firms, we also estimate an event study version of equation (4):

$$\log \text{College Salary}_{it} = \mu_i + \eta_t + \sum_{k=2008, k \neq 2013}^{2019} \delta^k D_i + \varepsilon_{it}. \quad (5)$$

4.2.2 Instrumental Variable Approach

The treatment approach in the previous section informs us about how the Swedish R&D reform directly affects Danish firms connected to Swedish firms, but it does not tell us how Danish R&D activity, proxied by salaries paid to college workers, responds to changes in R&D activity in connected Swedish firms. To examine this mechanism we adopt an instrumental variable approach similar to Kovak et al. (2021), that exploits the Swedish R&D reform.

Our specification of interest, the second stage, relates Danish firms' college salaries to the sum of college salaries across its trade and MNE partners according to the empirical model

$$\log \text{College Salary}_{it} = \alpha_i + \delta_t + \beta \log \sum_{j \in \Omega_i} \text{College Salary}_{jt} + \epsilon_{it}. \quad (6)$$

As defined above, Ω_i is the set of Swedish firms j that are linked to Danish firm i by a trade or MNE relationship. If the coefficient of interest $\beta > 0$, then Danish firms' R&D activity complements Swedish R&D activity. If the coefficient of interest $\beta < 0$, then Danish firms' R&D activity is a substitute for Swedish R&D activity.

The main identification challenge is simultaneity, i.e., if R&D activity across countries is a substitute or complement, then a shock to Danish R&D activity, ϵ_{it} will be systematically related to Swedish R&D activity across trade and MNE partners $\sum_{j \in \Omega_i} \text{College Salary}_{jt}$. We rely on Swedish firms' responses to the R&D reform to develop an instrumental variable and address this identification challenge.

The first step is to obtain predicted values for Swedish firms' college worker salaries

based on their exposure to the R&D reform following the approach in equation (??). That is, we compute predicted values for Swedish firms' college worker salaries the following way:

$$\log \widehat{\text{College Salary}}_{jt} = \hat{\mu}_j + \hat{\eta}_t + \hat{\beta}1\{t \geq 2014\} \times H_j. \quad (7)$$

We construct our instrumental variable by adding predicted college salaries for all Swedish firms connected to Danish firm i , i.e., $\sum_{j \in \Omega_i} \log \widehat{\text{College Salary}}_{jt}$. We then estimate equation (6) with 2SLS using this instrumental variable. That is, the first stage equation is

$$\log \sum_{j \in \Omega_i} \text{College Salary}_{jt} = \alpha_i + \delta_t + \beta \sum_{j \in \Omega_i} \log \widehat{\text{College Salary}}_{jt} + \epsilon_{it}. \quad (8)$$

The standard identification assumptions are that the instrument predicts the endogenous variable and is excluded from equation (6). We will address the first assumption in the results section. The second assumption plausibly holds, because variation in the predicted values $\log \widehat{\text{College Salary}}_{jt}$ is due to the Swedish policy reform and the Swedish policy reform was not designed based on the R&D performance of Danish firms.

5 Results for Swedish Firms

This section reports and discusses the estimation results for Swedish firms based on the data samples and identification strategies developed in Section 3 and 4. Across the various results, we emphasize estimation results for firms with trade and MNE links.

5.1 Effects of R&D reform on R&D expenditures

Using the information observed in the Swedish R&D survey (see Section 3.1), we estimate the OLS effect of the Swedish R&D reform on employment-based R&D activity according to the specification in equation (1). Table 4 reports the estimation results.

Table 4: Effects of R&D Reform on R&D Expenditures

Outcome (log):	R&D labor costs			R&D expenditures		
	(1)	(2)	(3)	(4)	(5)	(6)
$1\{t \geq 2014\}H_j$	0.15*** (0.059)	0.17*** (0.058)	0.17*** (0.063)	0.24*** (0.057)	0.24*** (0.058)	0.25*** (0.062)
$\dots \times MNE_j$			-0.014 (.068)			-0.011 (0.073)
Firm sample	All	Any link	Any link	All	Any link	Any link
Observations	3,599	3,344	3,344	3,664	3,402	3,402

Notes: Firm and year fixed effects included in all specifications. Description of estimation samples; All: Firms are present in the biennial R&D survey prior to the reform (2011 and/or 2013), and firms report positive R&D labor costs (in col. 1-3) or R&D expenditures (in col. 4-6) according to the biennial R&D survey (2011, 2013, 2015, 2017, 2019). Any link: Firms have a trade and/or MNE relationship with at least one Danish firm prior to the R&D reform (2012-2013). Robust standard errors in parentheses.

Table 4, column 1 shows the baseline estimation results over the observed Swedish R&D Survey estimation sample. According to the estimate, we find that firms affected by the R&D reform, i.e. $1\{t \geq 2014\} \times H_j = 1$, see a 15 percent increase in employment-related R&D expenses.

Column 2 focuses only on firms in the estimation sample that have a trade and/or MNE link with a Danish firm. The estimated effect of the R&D reform increases to 17 percent but is similar to the estimate reported in column 1. This is not surprising. The observation counts in columns 1 and 2 show that imposing the restriction that firms need to have a trade and/or MNE link with Danish firms does not substantially affect the sample.

Column 3 introduces an interaction effect to the baseline empirical model to examine if Swedish firms with an MNE link in Denmark see different effects compared to those firms with a non-MNE trade link in Denmark according to equation (2). The estimates imply that we can not reject the hypothesis that the interaction effect is different from zero at

conventional levels of significance. Therefore, we do not find evidence that Swedish firms with trade links were affected differently than Swedish firms with MNE links by the R&D reform.

Columns 4-6 repeats the approach from columns 1-3 but replaces the outcome variable with any kind of R&D expenditure. The reform effect is somewhat larger at 24 percent suggesting that higher R&D employment expenditures go hand in hand with other non-employment R&D expenditures such as equipment used for R&D activities.

An advantage of using the R&D survey is that we observe self-reported R&D activity. A disadvantage is that we do not observe mechanisms by which the R&D reforms may have affected R&D activity. This makes it challenging to examine the transmission of R&D activity across trade and MNE networks. To address this challenge, the next subsection examines the effect of the R&D reform on salaries paid to workers with college education.

5.2 Effects of R&D reform on College Salaries

In this subsection, we examine if the Swedish R&D reforms affected employment for workers potentially engaged in R&D activity. Using the information observed in the Swedish register data (see Section 3.2), we estimate the OLS effect of the Swedish R&D reform on (log) college salaries. Table 5 reports the results.

Table 5: Effects of R&D Reform on Salaries

Outcome (log):	College salaries			Wage bill	STEM sal.	PhD sal.
	(1)	(2)	(3)	(4)	(5)	(6)
$1\{t \geq 2014\}H_j$	0.16*** (0.026)	0.14*** (0.027)	0.15*** (0.029)	0.042 (0.028)	0.16*** (0.027)	0.24*** (0.042)
$\dots \times MNE_j$			-0.046 (0.054)			
Firm sample	All	Any link	Any link	Any link	Any link	Any link
Method	OLS	OLS	OLS	OLS	OLS	PPML
Observations	97,079	62,838	62,838	67,921	50,811	24,179

Notes: Firm and year fixed effects included in all specifications. Description of estimation samples; All: Firms are present in the biennial R&D survey prior to the reform (2011 and/or 2013), and firms have at least one employee with a college degree (2008-2019). Any link: Firms have a trade and/or MNE relationship with at least one Danish firm prior to the R&D reform (2012-2013). Robust standard errors in parentheses.

The estimate in Table 5, column 1 shows that the R&D reform increased college salaries in firms treated by the R&D reform. We consider this evidence that the R&D reform affected firms' R&D activity through increased employment/payment to college-educated workers.

Compared to the estimated effect of the R&D reform on R&D activity according to the Swedish survey data (Table 4, column 1) the effect on college salaries is similar in magnitude but improves in precision. Overall, our takeaway is that college salaries paid by Swedish firms are a reasonable proxy for firms' broader R&D activity consistent with firms' self-reported R&D activity, and the larger sample of the Swedish register data compared to the R&D survey provides more conclusive evidence on the mechanisms and the overall effects of the R&D reform.

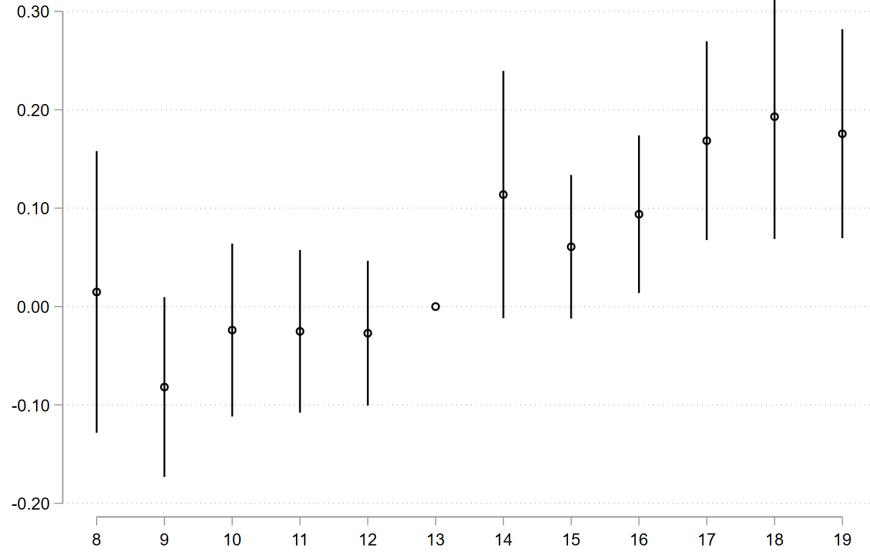
Table 5, column 2 focuses on only those Swedish firms that are part of a trade or MNE network with Danish firms. Although in this case, we see a substantial drop in the sample, the coefficient estimates and the standard errors remain roughly unchanged.

Column 3 presents estimates to examine if firms that are engaged in MNE networks with Danish firms respond differently to the R&D reform than those firms engaged in a trade network with Danish firms. The estimates do not provide evidence that firms engaged in MNE networks respond differently to the R&D reform in terms of its effect on college salaries.

Column 4 shows that there is no significant effect on the overall wage bill of the firm. This means that the treated firms become more intensive in the use of college educated workers. Column 5 considers the subset of college educated workers with a STEM degree who may be more directly involved in R&D activities in private firms. There is a positive reform effect on STEM worker salaries of the same order of magnitude as for college salaries. Finally, column 6 zooms in on PhD salaries. Since many firms do not hire workers with PhD degrees we estimate this effect using PPML. Again, we find that the reform boosts salaries for this group of presumably research oriented workers.

To check if treated firms were on a different trajectory already prior to the reform and if post reform effects persist, we estimate an event study version of the model according to equation (3) with (log) college salaries as outcome. Figure 1 presents the results and shows no evidence of differences in pre-trends. These results support the validity of our identification strategy. Furthermore it is evident that the treatment effects show persistency and even tends to increase to close to 20 percent in the last three post reform years, 2017-2019.

Figure 1: Effects of R&D Reform on College Salaries in Sweden



Notes: Estimation sample is similar to that of column 1 of Table 5, see table notes for details.

Based on the results presented in Table 4 and 5 and Figure 1 we conclude that the Swedish R&D reform increased R&D activity in firms that were (potentially) treated by the reform. Furthermore, based on college salaries and firms' self-reported R&D activity, the effect of the reform is similar for firms engaged in trade and MNE networks with Danish firms.

5.2.1 Extensions

Table 1 showed that most firms in our sample do not have expenses on R&D employment in the pre-reform years. This raises the question if the treatment effects are driven by firms that already had R&D employment expenses pre-reform, if they are driven by firms starting to use R&D workers or both. Table 6 repeats the approach from Table 5 but conditions the sample on firms that reported positive R&D employment expenses in the pre-reform years (either 2011 or 2013). Clearly, the sample size shrinks considerably, but we still find a treatment effect on college salaries of almost the same magnitude as for the full sample. This

suggests that the intensive margin contributes to the overall effect as firms raise expenses on college salaries. We now find a statistically significant effect on the overall wage bill (column 4), but this effect is still smaller than the effect on college salaries suggesting that research intensity rises in the treated firms. Finally, columns 5 and 6 again find positive effects on salaries for STEM workers and workers with a PhD degree.

Table 6: Effects of R&D Reform on College Salaries - Intensive Margin

Outcome (log):	College salaries			Wagebill	STEM sal.	PhD sal.
	(1)	(2)	(3)	(4)	(5)	(6)
$1\{t \geq 2014\}H_j$	0.13*** (0.028)	0.13*** (0.028)	0.15*** (0.031)	0.066** (0.030)	0.14*** (0.029)	0.24*** (0.042)
$\dots \times MNE_j$			-0.088 (0.057)			
Firm sample	All	Any link	Any link	Any link	Any link	Any link
Method	OLS	OLS	OLS	OLS	OLS	PPML
Observations	15,257	13,507	13,507	13,623	13,086	9,614

Notes: Firm and year fixed effects included in all specifications. Description of estimation samples; All: Firms report positive R&D labor costs prior to the reform (2011 and/or 2013), and firms have at least one employee with a college degree (2008-2019). Any link: Firms have a trade and/or MNE relationship with at least one Danish firm prior to the R&D reform (2012-2013). Robust standard errors in parentheses.

Firms below but close to the cap will only have the marginal costs of employing R&D workers reduced up to the cap, and so they stand to gain little compared with firms further from the cap when increasing R&D labor costs. This raises the question if firms distant from the cap drive the treatment effects estimated above. To examine this, we estimate an adjusted version of the model, where the treatment variable, H_j , takes the value 0 if firms are at or above the cap as before. For firms below the cap the treatment variable is 1 minus the ratio of pre-reform labor costs to the cap. That is, firms below but close to the cap now has a reduced contribution to the estimated effect compared with firms far from the cap.

Table 7 columns 1-2 shows that this adjustment to the model still produces significant

treatment effects, but they are smaller than before. This suggests that firms far from the cap also reacts to the reform, but firms closer to the cap appear to contribute with a stronger response. Columns 4-6 indicates that firms still increase salaries for STEM workers and workers with a PhD degree, but they now reduce the overall wage bill. We take the results of Table 6 and 7 as evidence that firms below the cap reacts to the reform irrespective of whether they are close to or far from the cap.

Table 7: Effects of R&D Reform on College Salaries - Distance to Cap

Outcome (log):	College salaries			Wagebill	STEM sal.	PhD sal.
	(1)	(2)	(3)	(4)	(5)	(6)
$1\{t \geq 2014\}H_j$	0.087*** (0.023)	0.066*** (0.024)	0.041 (0.027)	-0.056** (0.025)	0.092*** (0.025)	0.22*** (0.045)
$\dots \times MNE_j$			0.024 (0.049)			
Firm sample	All	Any link	Any link	Any link	Any link	Any link
Method	OLS	OLS	OLS	OLS	OLS	PPML
Observations	97,079	62,838	62,838	67,921	50,811	24,179

Notes: Firm and year fixed effects included in all specifications. Description of estimation samples; All: Firms are present in the biennial R&D survey prior to the reform (2011 and/or 2013), and firms have at least one employee with a college degree (2008-2019). Any link: Firms have a trade and/or MNE relationship with at least one Danish firm prior to the R&D reform (2012-2013). The treatment variable takes the value 0 if firms are at or above the cap. For firms below the cap, the treatment variable is defined as $H_j = 1 - \min\left(1; \frac{\text{pre-reform R\&D labor costs}}{\text{cap}}\right)$. Robust standard errors in parentheses.

In a final extension we examine if the R&D reform just boost hourly wage rates without changing the employment of R&D workers. This may happen if R&D workers are in fixed supply in the short and medium term, see e.g. Goolsbee (1998). That is, we use employment of college workers as outcome instead of annual salaries and find similar sized elasticities (see Appendix Table A3). For example, while the estimated elasticity for college salaries in column 2 of Table 5 is 0.14, the corresponding elasticity for the number of college workers is 0.12 suggesting that most of the reform effect in fact increases R&D activities in the treated

firms. We take this as evidence that the R&D reform indeed increased R&D activity of the treated firms.

6 Spillover Effects

We now turn to how Danish firms that are linked to Swedish firms through trade or MNE ownership are affected by the Swedish R&D reform. We first use our reduced form approach outlined in Section 4.2.1. We then provide evidence for mechanisms, and finally we present results from the instrumental variable approach explained in Section 4.2.2.

6.1 Reduced Form Spillover Effects

In this subsection, we examine if Danish firms that are linked to Swedish firms via trade or MNE networks are affected by their Swedish partners' R&D activity. In particular, we examine if, among the Danish firms that are linked to Swedish firms, those firms whose partners were potentially affected by the R&D reform adjust their own R&D activity based on salaries paid to college workers in Denmark (see Section 3.5 for details) and equation (4). We estimate the model using OLS and Table 8 reports the estimation results.

Table 8: Reduced Form Effects of R&D Reform on Danish Firms

Outcome (log):	College salaries			Wage bill		
	(1)	(2)	(3)	(4)	(5)	(6)
$1\{t \geq 2014\}D_i$	0.061*	-0.20***	0.095***	-0.0097	-0.24***	0.0017
	(0.034)	(0.056)	(0.035)	(0.028)	(0.053)	(0.028)
Firm sample	Any link	MNE	Trade only	Any link	MNE	Trade only
Observations	103,148	8,538	101,187	103,148	8,538	101,187

Notes: Firm and year fixed effects included in all specifications. Description of estimation samples; Any link: Firms have a trade and/or MNE relationship with at least one Swedish firm prior to the R&D reform (2012-2013), and firms have at least one employee with a college degree (2008-2019). MNE: Firms have at least one Swedish MNE relationship prior to the R&D reform (2012-2013). Trade only: Firms have at least one Swedish trade relationship that is not also an MNE relationship prior to the R&D reform (2012-2013). Robust standard errors in parentheses.

The estimate in Table 8, column 1 provides evidence that R&D activity by Swedish firms affects their Danish partners' R&D activity. In particular, Danish firms with trade and/or MNE partners in Sweden who are potentially affected by the R&D reform increase their own R&D activity by about 6 percent.

To examine if firms in MNE and trade networks respond differently to foreign R&D shocks, we focus on the sample of Danish firms with MNE links in Sweden (column 2) and Danish firms that are connected to Swedish firms by non-MNE trade links (column 3).¹⁶ The estimate reported in column 2 provides evidence that within MNE networks R&D activity is a substitute, i.e., Danish firms that are linked to Swedish MNEs that are treated by the Swedish R&D reform see a decrease in their own R&D activity. That is, a R&D reform induced increase in college salaries in Swedish firms reduces college salaries in Danish affiliates of the same MNE. This is opposite to the positive correlation in college salaries between affiliates of the same MNE in the raw data documented in Appendix Figure A3 and illustrates the importance of using a plausibly exogenous change in R&D employment

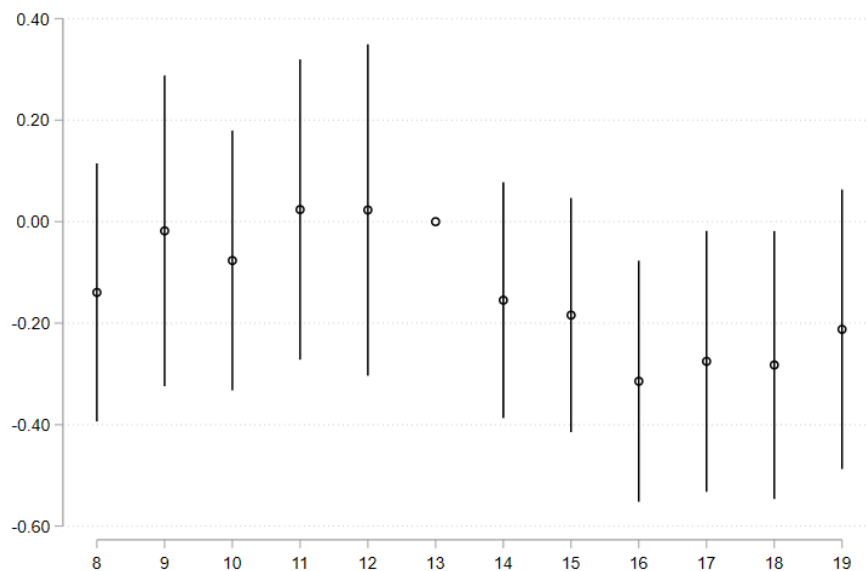
¹⁶Note that these two groups are not mutually exclusive since a Danish firm could have an MNE relationship with one Swedish firm and a non-MNE trade relationship with another.

to uncover spillover effects within MNE networks.

In contrast, the estimate in column 3 shows that within trade networks R&D activity is a complement, i.e., Danish firms that trade with Swedish firms, that are treated by the R&D reform, see an increase in their own R&D activity as a consequence of the reform. This is evidence for positive R&D employment spillover effects, that can potentially justify subsidizing R&D activities.

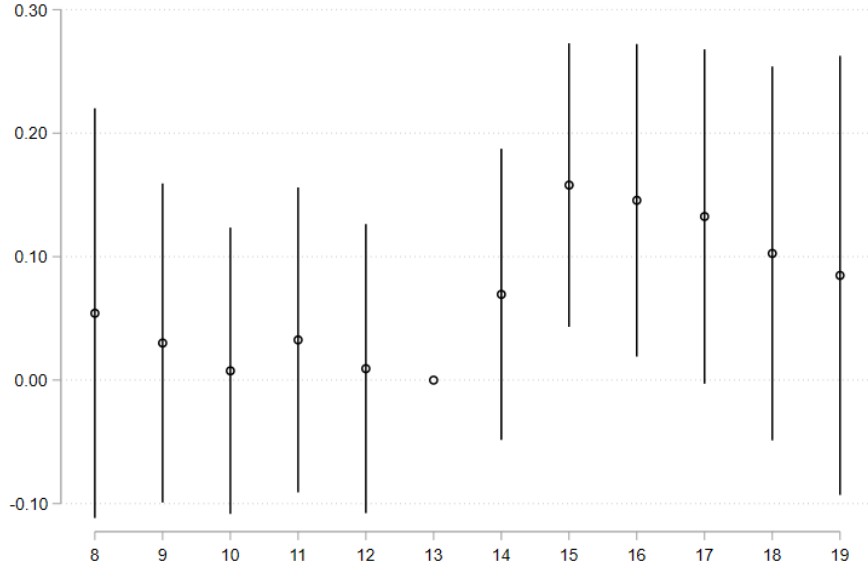
As in the previous subsection, we investigate effect persistency and if potentially treated Danish firms were on a different trajectory than other Danish firms already prior to the reform. We do so by estimating the event study version of the model in equation (5) and plot the results in Figure 2 for MNE linked firms and in Figure 3 for trade linked firms. As above, there is no evidence of significant differences in pre-trends but the effects tends to fade over time such that they are no longer significantly different from zero four to five years after the reform.

Figure 2: Effects of R&D Reform on College Salaries in Danish MNE linked firms



Notes: Estimation sample is similar to column 2 of Table 8, see table notes for details.

Figure 3: Effects of R&D Reform on College Salaries in Danish trade linked firms



Notes: Estimation sample is similar to column 3 of Table 8, see table notes for details.

6.2 Mechanisms

This subsection zooms in on the negative reform effect on R&D activity in Danish affiliates of the same MNE as the treated Swedish firms as shown in Table 8. Treated Swedish firms hire college educated workers while MNE-linked Danish firms reduce employment of college workers. One possible mechanism is that the Swedish firms simply hire workers from the Danish affiliates. This is not unlikely since language and culture is similar in the two countries and local labor markets have increasingly been integrated since the opening of the Øresund Bridge connecting Copenhagen and Malmö in 2000, see, e.g., Bütikofer et al. (2024).

To investigate this channel we exploit that we can track workers across the Danish and Swedish borders. That is, we observe if a Swedish firm hires a worker employed by a Danish firm the same or the previous year. We observe 948 such hires in the sample of Swedish firms with any type of link to a Danish firm, but once we restrict it to the sample with

MNE linked Swedish firms, we have only 382 such hires over the period 2009-2019. Since it is a rather rare event that Swedish firms hire workers from Danish connected firms, the data becomes too thin to examine hiring of the subgroup of college educated workers. From these observations we construct as outcome variables dummies at the firm-year level for hiring from a Danish linked firm, and we then use our reduced form approach from above to estimate the effect of the R&D reform on hiring from Danish connected firms.

Column (1) of Table 9 shows that in the sample of Swedish firms with any type of link (trade or MNE), the reform increased the probability of hiring a worker from a Danish linked firm by 2.2 percent more than for non-treated firms. However, column (2) shows that when restricting the sample to Swedish firms with only trade links to Danish firms, there is no significant effect on hiring from a Danish linked firm. By contrast, when restricting the sample to Swedish firms linked to a Danish firm through MNE ownership, there is a marginally significant effect, see column (3). Thus, the overall effect in column (1) appears to be driven by MNE linked firms. Finally, column (4) shows that there is strongly significant effect on hiring, when adjusting the outcome variable to be a dummy for hiring from a Danish firm with both trade and MNE links. Overall, this suggests that MNEs appear to hire workers from their affiliates abroad when employment subsidies rise at home. We take this as suggestive evidence only, since the hires we examine are not narrowed down to college educated workers but includes any kind of worker previously employed by a connected Danish firm.

6.3 IV Spillover Effects

While the reduced form approach in Section 6.1 is a direct way of assessing the impact on R&D employment in international firm-to-firm networks, it is difficult to evaluate the *extent* to which Danish and Swedish members of the same trade or MNE networks substitute R&D activity.

This section presents instrumental variable estimates that clarify these relationships. That is, it reports elasticity estimates for Danish college worker salaries with respect to

Table 9: Cross-Border Hires

Outcome (dummy):	Linked hires	Trade linked hires	MNE linked hires	MNE and trade linked hires
$1\{t \geq 2014\}H_j$	0.022*** (0.0082)	0.0100 (0.0066)	0.020* (0.011)	0.026*** (0.010)
Sample	Any link	Trade only	MNE	MNE
Mean outcome	0.011	0.0066	0.017	0.011
N	62,238	60,095	15,163	15,163

Notes: Firm and year fixed effects included in all specifications. Description of estimation samples; Any link: Firms have a trade and/or MNE relationship with at least one Swedish firm prior to the R&D reform (2012-2013), and firms have at least one employee with a college degree (2008-2019). MNE: Firms have at least one Swedish MNE relationship prior to the R&D reform (2012-2013). Trade only: Firms have at least one Swedish trade relationship that is not also an MNE relationship prior to the R&D reform (2012-2013). Robust standard errors in parentheses.

R&D-related employment in linked Swedish firms. We consider three types of estimates. First, as a benchmark, we estimate the relationship between Danish firms' salaries of college workers and the total sum of their Swedish partners' R&D activity according to equation (6) with OLS. Second, we re-estimate equation (6) applying the instrumental variable strategy discussed in Section 4.2.2. Finally, we will relate Danish college workers' salaries directly to the instrumental variable constructed in Section 4.2.2 in a reduced form exercise. Table 10 reports the results.

Table 10: IV Effects of Swedish R&D on Danish Firms

Outcome: log College salaries	Long panel (2008–19)			Short panel (2010–16)		
	Any link	MNE	Trade only	Any link	MNE	Trade only
<i>Panel A: OLS</i>						
$\log \sum_{j \in \Omega_i} y_{jt}$	0.040*** (0.0065)	0.047** (0.020)	0.041*** (0.0066)	0.035*** (0.0097)	0.025 (0.023)	0.038*** (0.0098)
<i>Panel B: IV</i>						
$\log \sum_{j \in \Omega_i} y_{jt}$	0.20*** (0.046)	-0.20 (0.15)	0.21*** (0.044)	0.21*** (0.073)	-0.38** (0.18)	0.26*** (0.072)
<i>Panel C: Reduced form</i>						
$\sum_{j \in \Omega_i} \widehat{\log y_{jt}}$	0.00099*** (0.00023)	-0.0011 (0.00084)	0.0011*** (0.00025)	0.0010*** (0.00036)	-0.0032** (0.0014)	0.0013*** (0.00037)
First-stage F-stat	648	124	545	309	41	298
Observations	103,148	8,538	101,187	62,926	5,241	61,697

Notes: Robust standard errors in parentheses.

Panel A provides the OLS results for the long panel used above (2008-2019) and for a shorter panel spanning 2010-2016. For the long panel, the coefficient estimates reported in columns 1, 2, and 3 all suggest that R&D activity within trade and MNE networks are complements, i.e., an increase in R&D activity as proxied by college workers salaries in Sweden results in an increase in college worker salaries in Denmark across all (column 1), MNE (column 2), and trade linkages (column 3). This is also true for the short panel, except for the MNE coefficient estimate which is no longer significant.

Panel B reports instrumental variable estimates and the conclusions are markedly different.¹⁷ The estimate in Column 1 is positive but not significant. The positive sign suggests that estimated across all links, R&D activity within trade and MNE networks is complementary across network members.

Columns 2 and 5 focuses on the sample of MNE linked firms. The effect is negative in the long panel but insignificant. By contrast, in the short panel there is a significantly

¹⁷For all the instrumental variable estimates, the instrument is predictive and not weak based on standard test statistics.

negative effect, which is consistent with the fact that the reduced form effects fade away over time, see Figure 2. According to the estimate, a one percent increase in total R&D activity across all Swedish MNE partners leads to a 0.4 percent decrease in Danish firms' salaries paid to college workers. Therefore, within MNEs R&D activity is a substitute.

Columns 3 and 6 focus on trade links, and the coefficient estimates are positive showing that R&D employment is complementary across firms within the same international trade network.

In summary, the results of Panel B, columns 1 to 3 are in line with the reduced-form evidence reported in Table 8. Pooling across trade and MNE links results in estimates that suggest that R&D activity is complementary across members of trade and MNE networks. However, leveraging our data we are able to distinguish MNE from trade networks and the evidence shows that although R&D activity is a complement across members of a trade network, it is a substitute across firms that are part of an MNE network.

Panel C reports the results when we relate log college worker salaries directly to the instrumental variable. Reassuringly, the estimates follow a similar pattern as the instrumental variable estimates reported in Panel B.

We interpret the instrumental variable estimates in Panel B as the causal effect on a Danish firm's R&D activity from a change in the combined R&D activity by its Swedish counterparts (where R&D activity is proxied by salaries paid to college workers). To get a better sense of the magnitudes, we use the estimates to perform a rough calculation of the effect of the Swedish R&D reform on the number of college workers in Swedish and in the Danish MNE counterparts, respectively. First, recall from Table 5 that the reform increased R&D activity by about 14 % for firms below the deduction cap prior to the reform. Since Swedish firms with Danish MNE relationships paid 36 million SEK in college salaries, this amounts to an increase of about 5 million SEK, which, by applying the average yearly wage, corresponds to an increase of about 10 college workers. How large is the effect on Danish MNE counterparts in comparison? Applying the estimated elasticity of -0.4, a 10% increase in R&D activity in a Swedish firm is accompanied by a 4% drop for the Danish

MNE counterpart. Based on average values of salaries paid and annual wages for the Danish estimation sample, this 4% drop corresponds to almost four college workers.

If we instead focus on Swedish and Danish firms connected through trade relationships, the reform incentivizes Swedish firms to hire about five additional college workers, which translates into an increase in college employment of about 0.2 for the average Danish counterpart.

7 Conclusions

In this articles we combine several data sources to examine the allocation of R&D activity within trade and MNE networks. We take advantage of a quasi-random policy experiment that incentivizes R&D activity in Swedish firms. Linking Swedish firms to Danish firms via trade and MNE networks, we then estimate substitution patterns in R&D activity among MNE and trade network members. Our data and identification strategy provides evidence that while R&D activity is a strategic complement among trade networks members, it is a strategic substitute among MNE members.

These results may reflect that when firms receive employment related R&D subsidies, they increase R&D activities and this spills over to trade linked partners, that also increase activities, perhaps due to knowledge spillovers, increased demand in the trade network or both. Do these positive spillovers justify R&D subsidies? On the one hand, the Swedish government may not value potential benefits to Danish firms. On the other hand, given the documented positive cross border spillovers, there will likely also be positive spillovers to trade linked domestic firms.

The negative spillover effect in MNE linked firms may reflect that MNEs reallocate tasks within the corporation such that R&D activities are shifted to the location with lowest marginal costs. That is, MNEs in position to move R&D activities to countries offering the most generous R&D subsidies not unlike how they also are known to move profits to countries with low corporate tax rates. Since MNEs account for the majority of R&D

activities in our data, this substitution effect is important, and it raises the question if the rising R&D subsidy rates in OECD countries have reached a too high level.

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Appendices

A Additional Tables and Figures

Table A1: Summary Statistics for Regression Samples

	Sweden			Denmark	
	All	Trade only	MNE	Trade only	MNE
#Firms	9,998	5,876	1,511	10,365	885
Industry affiliation (%)					
- C: Manuf.	22.3	32.3	29.8	24.9	18.9
- G: W/R	12.8	16.9	15.7	32.6	32.0
- H: Transp.	10.3	7.3	5.7	5.3	6.8
- J: Info/comm.	8.3	9.4	13.0	6.3	12.7
- M: Prof. act.	15.4	13.8	12.0	7.7	9.3
- Other	31.0	20.4	23.8	23.3	20.5
#Employees					
- Mean	148.0	210.2	395.7	213.9	406.2
- Median	29.0	37.7	76.8	33.0	66.0
College Workers					
- Share with college workers	90.8	94.1	97.6	92.9	96.3
- College wage bill share among > 0	33.6	33.2	38.3	32.4	36.2
STEM workers					
- Share with STEM workers	69.4	77.9	88.1	71.2	81.5
- STEM wage bill share among > 0	16.0	16.6	17.3	16.3	15.3
PhD workers					
- Share with PhD workers	17.9	23.6	33.8	13.8	21.7
- PhD wage bill share among > 0	7.0	6.5	4.8	4.4	2.7

Notes. The samples include.... The distribution of firms is measured in pre-reform years... Industries are classified according to... STEM requires college degree.

Table A2: Trade Patterns for Swedish Firms

	Export	Import	MNE
Share of firms trading with EU / part of MNE	49	76	37
... of which, share of firms with DK relationship	72	65	41
... of which:			
share of value to/from DK	31	25	
share of firms with DK as top market in terms of value	33	25	
share of relationships in DK	31		11
share of firms with DK as top market in terms of relationships	49		10

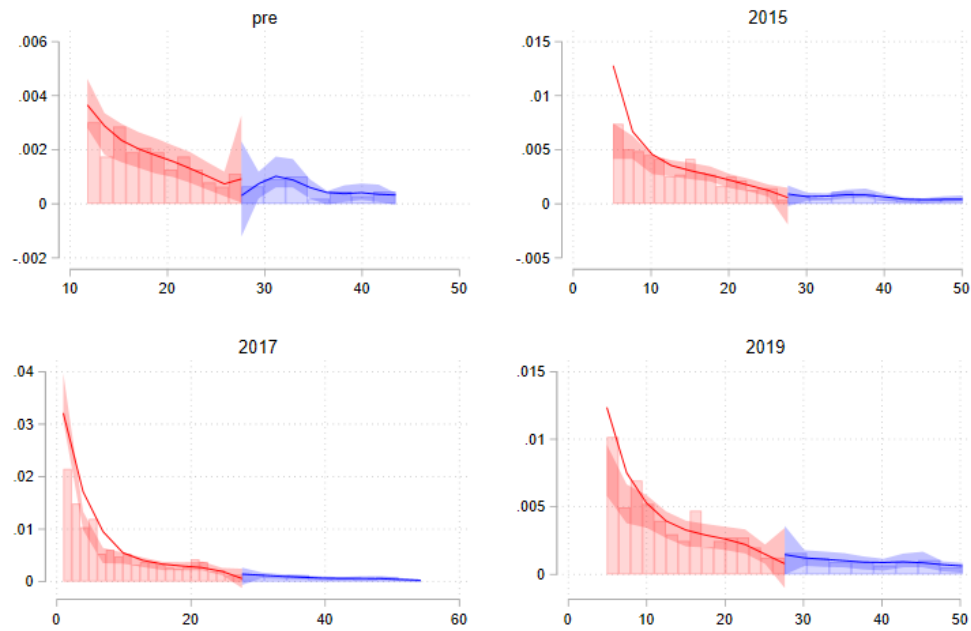
Note: Trade data (VIES) cover within-EU trade only. MNE data cover all global relationships, including domestic (but firms with only domestic relationships are not considered part of an MNE). Measured pre-reform.

Table A3: Effects of R&D Reform on Employment

Outcome (log):	College workers			All workers	STEM workers	PhD workers
	(1)	(2)	(3)	(4)	(5)	(6)
$1\{t \geq 2014\}H_j$	0.13*** (0.024)	0.12*** (0.024)	0.13*** (0.030)	0.052** (0.026)	0.12*** (0.024)	0.25*** (0.039)
$\dots \times MNE_j$			-0.042 (0.048)			
Firm sample	All	Any link	Any link	Any link	Any link	Any link
Method	OLS	OLS	OLS	OLS	OLS	PPML
Observations	97,079	62,838	62,838	67,921	50,811	24,179

Notes: Firm and year fixed effects included in all specifications. Description of estimation samples; All: Firms are present in the biennial R&D survey prior to the reform (2011 and/or 2013), and firms have at least one employee with a college degree (2008-2019). Any link: Firms have a trade and/or MNE relationship with at least one Danish firm prior to the R&D reform (2012-2013). Robust standard errors in parentheses.

Figure A1: Distribution of R&D employment expenditures for Swedish firms



Notes: Sample: surveys etc.

Figure A2: Wage Bill Correlation between Swedish and Danish firms

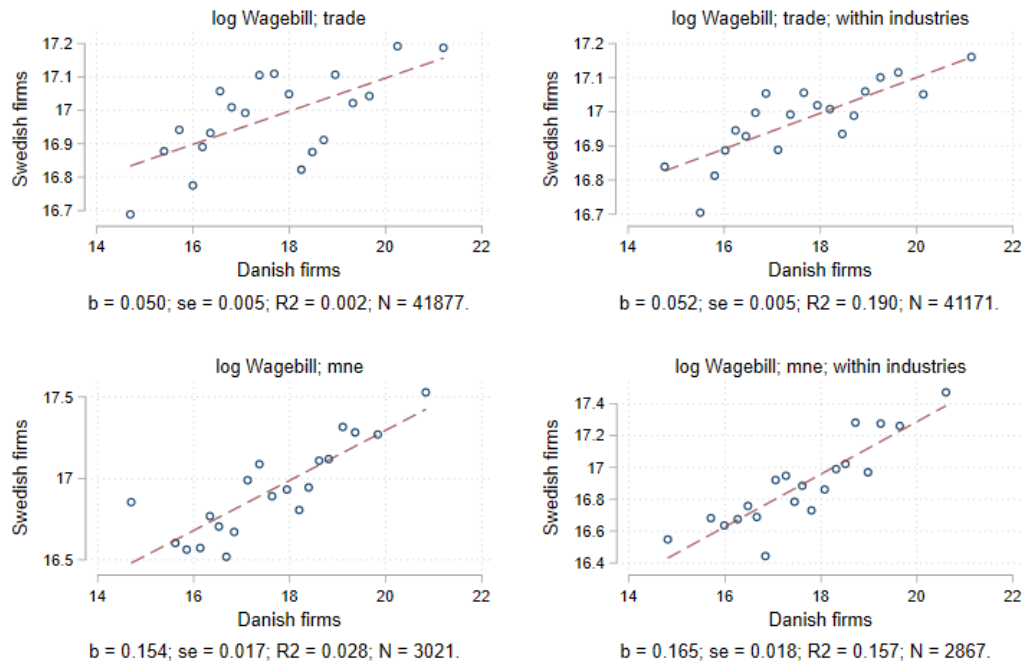
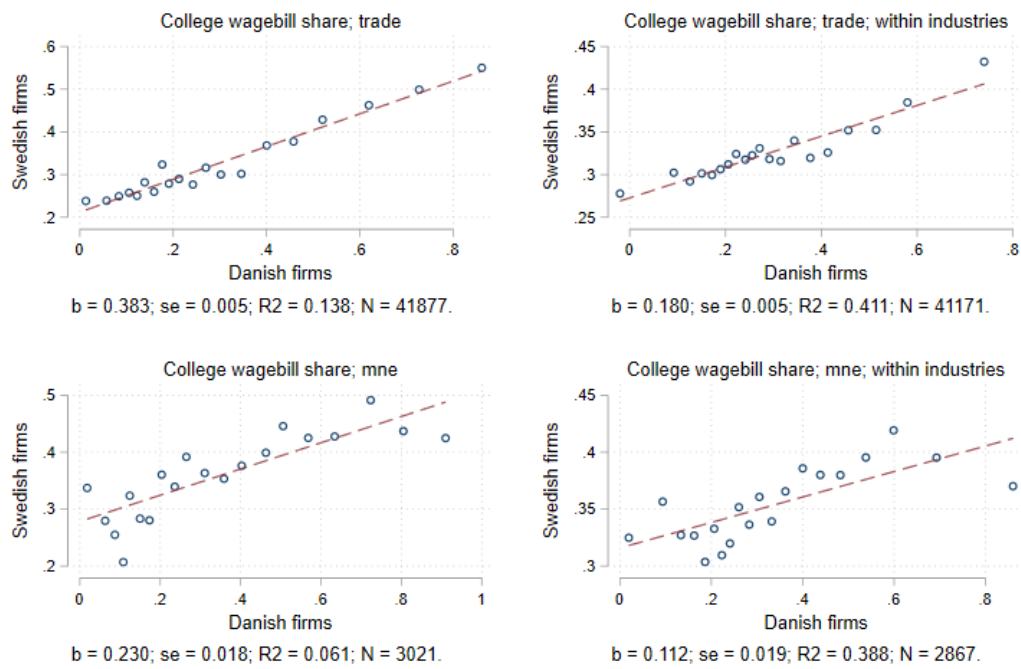


Figure A3: College Worker Wage Bill Correlation Between Swedish and Danish Firms



B Construction of Firm-to-Firm Ownership Links

This section describes the construction of extensive firm-to-firm ownership linkages between firms in Sweden and Denmark.¹⁸ By merging a Swedish group register from Bisnode with a Danish firm ownership register from Experian, we construct a dataset containing all ownership linkages involving a Danish or Swedish firm along with information on the linkages’ underlying firm-groups.

Both the Bisnode and Experian datasets include information on the group structure. Both datasets include domestic firm identifiers but lack an official firm ID for foreign firms. To construct a firm-to-firm matched data that can be linked to other microdata, we must add official identifiers from other sources. This appendix describes how that match is conducted. The overall match rate is high, and we are able to find a Danish firm identifier for 98.8% of the Danish firms in the Swedish data and 91% of the Swedish firms in the Experian data.

B.1 Bisnode: The Swedish Group Register

The first dataset is the Swedish Group Register collected by the data analytics company Bisnode (now part of Dun and Bradstreet). Groups need to submit consolidated accounts (Swedish: Koncernredovisning) that encompasses the group mother and all the subsidiary daughter firms as if they were a single economic unit. In 1996, Bisnode started to manually collect this data and updated yearly since. This data is of high quality and used for example to form the population of international firms and foreign ownership by Statistics Sweden (SCB). The group register includes information on the global business group structure of all groups that include at least one Swedish firm. A “Group” is an ownership structure between legally distinct firms that are under common financial management. For a firm to be included in a group, it must be under majority control by the group mother (directly or indirectly through ownership chains).

In the Swedish Group Register, a “group” consists of one group mother, and then all subsidiary firms which are ultimately fully, or partly, owned by the group mother. The

¹⁸A more detailed description and the accompanying code are available on request.

Group Register includes, among other variables, names and firm ID numbers (official Swedish “organisationsnummer”) of all firms located in Sweden. However, when a firm in the group is located outside of Sweden the same variables are included while the firm ID is in those cases a serial-number. Hence, for foreign (Danish) firms in the Group Register the firm ID does *not* consist of the official foreign firm ID (for Danish firms the ID is called a CVR number). To construct firm-to-firm ownership linkages between Denmark and Sweden, that can be linked with other firm level data, we need to add the official Danish firm ID number (CVR) to the Group Register. Below, we document how we add these identifiers to the data.

B.2 Structure of the Data

The Bisnode Group Register describes the group structure of all groups where Swedish firms are involved, the period in which these firms are included in the group, the ownership percentage and which country the firm is from. Since, a firm needs to be majority controlled to be included in a group, a single firm can only be in one group at any point in time.¹⁹ The ownership structure is organized through chains of mother-daughter firm relations, in which the mother firm owns all or part of the daughter firm. There is only one group mother in any group, but there can be multiple subsidiary/intermediate mother firms. Hence, the group mother is not necessarily the direct mother to all firms in the group but rather the mother of the intermediate mother firm.

Each observation in the register includes a firm ID specific to the firm in question (official firm ID if a firm is Swedish and a serial number otherwise). Each observation is identified by the combination of the firm ID variable ‘od12’ and year. For each observation, firm-year, we can also see the firm ID of the mother firm that directly owns the firm (variable om12) as well as the firm group ID (variable ok12) of the ultimate group mother. From these three

¹⁹However, in the Group Register we do observe a small share of firms with multiple observations in the same year. This can stem from firms changing their accounting period in a year or the selling/buying of firms when the accounting periods do not perfectly overlap. Hence, if a firm has been sold after the accounting period ends of the group selling but before the end of the group buying, the same firm can have multiple observations in a year. Around 1.2% of the observations (1% of the firms) relate to firms with multiple observations within a year. Of those observations, roughly 40% are related to firms that are a part of multiple groups under the same year. The remaining cases are related to firms with multiple accounting periods in a year.

firm ID variables, we can imply the relation between any two firms in a group through a chain of mother-daughter relations. When a firm is Swedish, an official firm identifier is used and a serial number for foreign firms.

B.3 The Matching Process

The goal is to connect all Danish firms in the Group Register with their corresponding CVR-number. We do this by matching the names in the Group register with those found in the public Central Business Register (CVR - Det Centrale Virksomhedsregister). As there may be discrepancies in the naming, this section describes the matching process. Over 90% of this problem is solved with minimal editing and generalizations of the names before merging (for example by translations, removing of punctuation marks etc.), but a smaller number of matches are found with other methods (manual matching or utilizing fuzzy name matching).

B.4 Reformatting of Data (Prepare_datasets.ipynb)

This section describes the process executed in the python scripts (saved as Jupyter Notebooks) Prepare_datasets.ipynb and matching.ipynb. To simplify the matching, the Bisnode and CVR-number datasets are reformatted to be more alike. This includes e.g.: 1) all letters are made to lowercase, 2) reformatting of language specific letters in the names are replaced with a common interpretation e.g. {ø:oe, æ:ae, å:aa, ö:oe, ä:ae, á:a, é:e, è:e, à:a, á:a}. 3) identifying and specifying different types of Danish firms (e.g. A/S, ApS). We reformat the CVR data to the same format as the Bisnode data (e.g. transform to yearly data).

B.5 Steps of the Matching Process (matching.ipynb)

The matching process is done in steps. At each step, the names of the firms are generalized a bit and then a merged. The merge is done using the merge_match()-function defined in the notebook. The matches are saved, while the unmatched names in the Group Register are moved to the next step. After all the steps of name generalizations and direct merging, fuzzy

name matching is done. At the end, a few remaining names are matched manually using <https://data.virk.dk/>. There are 9,613 unique od12-numbers in the raw data (some of them have multiple names), and we find a corresponding CVR for 9,533 of them. These numbers include minority owned firms and branches, which are excluded for the analysis of this paper.

Step 1 - Merge on exact name and type: First we merge on ‘name’ and firm type. This removes all the ‘exact matches’ where there are no discrepancies in the names. The `merge_match()`-function also only matches firms, if the CVR has been active in at least a subset of the active period in the Group Register. This step matches 74.8% of the Group Register names.

Step 2 - Removal of e.g. punctuation marks: In the second step, punctuation marks are removed, & is translated to the Danish word “og”, and “oe” and “aa” are generalized to “o” and “a”. Some examples of other changes include: international:int, and:og, av:af, of:af, limited:ltd, denmark:dk, danmark:dk, holdings:holding and sweden:se. Spaces are also removed in this step. This step matches 9.5% of the Group Register names.

Step 3 - Ignoring firm type: Match only on name and not on firm type. This step matches 2.9% of the Group Register names.

Step 4 With spacing, removal of recurrent words and common translations: Same as above but use spacing but remove recurrent words with little unique firm (e.g. ‘asset’, ‘assets’, ‘broderne’, ‘brdr’, ‘brodrerne’ ‘holding’). Some words are given a common translation. This step matches 3.5% of the Group Register names, meaning that 90.7% of the Group Register names have been matched at this point.

Step 5 - Fuzzy name matching. In the script the python package `fuzzywuzzy`²⁰ is used, but the process could also be implemented in other languages. The package uses the

²⁰<https://github.com/seatgeek/fuzzywuzzy>

Levenshtein distance²¹, which is a mathematical formula for calculating how different two strings are, to calculate a ratio of similarity from 0 to 100, for each remaining unmatched name in the Group Register data compared with all names in the CVR data (again first the names, then the binames). For each Group Register name, the highest ratio is chosen as the correct match, as long as it is above a certain threshold. The names which are compared are formatted in the same way as in the second step, but without removing spaces. This step matches 6.2% of the Group Register names.

Step 6 - Manual match <https://data.virk.dk/> and searching the internet. At this step 1.9% of the Group Register names are matched, while the remaining 1.2% are discarded.

Step 7 - Post-matching data checks: We perform a number of smaller checks on the data. See the online appendix for details.

B.6 Summary of Matching Process

The Group Register covers groups with a Swedish firm during the period 1995-2018. Even though our main aim is to investigate groups with a Danish connection, the register itself includes a much broader scope of foreign firms regardless of country of origin. Using the approach described above, **a match rate of 98.8% is achieved**. Of the matches, 74.8% was matched with minimal editing (only initial reformatting), while 15.9% are matched after some editing in steps 2-4. Hence, over 90% of are matched with minimal editing and are hence expected to be of high quality. The outcome is a list of the CVR-numbers of almost all Danish firms which are part of groups that includes a Swedish firm from 1995-2018.

C Experian: Danish Firm Ownership Data

Our goal is to create a dataset that includes all groups with a Danish or Swedish firm involved and where every Danish or Swedish ownership link is included. The Swedish Group

²¹https://en.wikipedia.org/wiki/Levenshtein_distance

Register includes comprehensive information on all groups where a Swedish company is involved. However, foreign companies (e.g. Danish) are generally only included in the data if they have a direct ownership-link to a Swedish firm. Hence, if a Danish group has multiple Danish firm and one of them is connected to Sweden, we may overlook such indirect ownership ties. For this, we need more information on links involving Danish companies. In appendix C we describe how we use data from Experian to solve this issue. The data from Experian includes the ownership structure of Danish groups from 1995 and forward.

The sections below describes how the Experian dataset is processed and made similar to Bisnode (add local firm identifiers). The work done in the python scripts (saved as Jupyter Notebooks) `Experian_clean_and_match.ipynb`, `Od12_key.ipynb`, `Third.country_key.ipynb`, `Experian_long.ipynb` and `Experian_final.ipynb`.

C.1 Cleaning Dataset (`Experian_clean_and_match`)

The Experian data is in a wide format, where one observation fully characterizes an ownership link. We have ‘RelationStart’ and ‘RelationSlut’ variables that indicate when the linkages started and ended, along with the ownership percentage. Furthermore, we have the name and country code for both companies involved, along with the CVR numbers for Danish companies. For Danish companies, we have another unique ID called ”KOB”. Prior to matching to Bisnode the data is cleaned and we remove observations where we have missing CVR numbers, names, missing both start and end dates. We remove missing values and add CVR numbers using step 1 and 2 above for a small number of observations (less than 2%).

C.2 Creating an ID Key for Swedish Companies in Experian (`Od12_key.ipynb`)

To fully cover all Danish-Swedish links, and to make sure we create a structural consistent dataset, we want to not only merge the datasets on Danish companies but also on Swedish companies when possible. Therefore, we have created a key that connects the Swedish companies in Experian with their organizations number from Bisnode.²²

²²Note that we do not have a comparable data source to public CVR register available to us. We hence rely on the firms in the Bisnode data to add Swedish local firm identifiers.

To do this, we have started with giving all Swedish companies in Experian a serial number. These serial numbers are given so that companies with the same name have the same serial number.²³ We then match the companies with Bisnode using three steps:

1. We try to match the companies by both name matching and by matching the Danish companies, they are linked to. Every Swedish company in Experian is linked to a Danish company, which we can utilize to ensure a correct match. We use the name matching procedure described in section B.5 but adds the constraint that a match also has to be linked to the same Danish company (same CVR number) in both datasets.
2. We match the remaining companies using only the connected Danish companies. For this step, we divide the Swedish companies into two categories, mothers and daughters. For mothers: If a specific Danish company in a given time frame has a Swedish mother company with an ownership share above 50% in both datasets, we take the organization number from the Swedish mother in Bisnode and assign it to the Swedish mother in Experian ²⁴. For daughters: If a Danish company owns only one Swedish company in both datasets around the same time period, we match the two daughter companies. If a Danish company owns multiple Swedish companies in a specific time period, we only match if the time periods are identical in both datasets.
3. The last remaining are companies where we have not been able to find the equivalent Danish link in Bisnode. For them, we match on names using the first two steps from the name matching procedure of section B.5. We also use a time restriction, saying that the time periods where the firms are in each dataset has to overlap with at least one year.

In total, we have managed to match 74% of the Swedish companies in Experian with an organization number from Bisnode. Around 53% of the Swedish firms in Experian have

²³Unfortunately, Experian sometimes have different observations involving the same company but where the company name is spelled slightly different. To get around this we have used a reformatting function that uncapitalized all letters, removes different signs and removes spacing.

²⁴Note that all majority links have an ownership share above 50%, so only one mother is possible for each daughter

been matched through the first step, 10% have been matched through the second step and 11% have been matched through the third step. Note that we remove minority owned firms for our analysis which tend to have lower match rates. The match rate for the sample of majority owned firms is considerably higher, around 91%.²⁵

D Restructuring Experian to the Bisnode Structure (Experian_long.ipynb)

In order to make the merge of the datasets easier, we want to restructure the Experian dataset so that it follows the Bisnode structure. To do this we firstly need to convert Experian into a long format where each ownership linkage has an observation in every year where it is active. We have the 'RelationStart' and 'RelationSlut' variables that indicate when the linkages started and ended. However, some observations have an empty 'RelationStart' variable, which by Experian's own explanation means that the relation is older than 2000. But as most of these linkages also end before 2000 ('RelationSlut' is before 2000) we have set the missing 'RelationStart' values to 1994 (the year the Bisnode data also starts). Out of a total of 570,000 useful observations in Experian, there are 20,000 of them that are missing Relation Start values. Linkages that ended earlier than 1994 are removed from the data. We then change Experian into a long format by creating an observation in every year between their 'RelationStart' and 'RelationSlut' years for a given ownership linkage. Secondly, we need to add information about the group for which the ownership linkage is connected to. Bisnode has a group mother variable that identifies the group, and a layer variable that states which layer of the group a given linkage lies in.²⁶

For the majority ownership links, we add group information by first identifying all mother companies in Experian that do not also have an observation as a daughter, i.e. they are not owned by another company in the data. These companies are thereby the data's group

²⁵The matches from the first and third step are very secure. For the third, we have been very conservative by only choosing to use the first two steps of the name-matching process in order to minimize wrong matches. In cases of disagreement, we rely on the Bisnode data.

²⁶The data includes minority ownership links, which we also refer to as interest links. These links offer the possibility that firms have multiple owners and thereby can be part of multiple groups. The problem now arises if these firms also are the owners of other firms. It now becomes non-obvious how to decide what groups these firms should be part of. We therefore decide not to add group information to any minority ownership links, which means that the group structures in our data are unique.

mothers. We then add the group-mother and layer variables to the data by iterating through linkages, starting with the group mothers and then marking all companies in the same ownership chain with the same group mother, and defining layer as the iteration number.

E Third Country Key (Third_country_key.ipynb)

Using Experian and Bisnode, it is not only possible for us to identify direct ownership linkages between Danish and Swedish companies. We also have the potential to identify companies that are linked through a common third country mother company. In these instances, we will have an ownership linkage with the Danish daughter in Experian and an ownership linkage with the Swedish daughter in Bisnode. To identify these linkages when merging the datasets, we need to make sure that the third country mother has the same (fake ID in both Experian and Bisnode. This is what the third country key is for. We use the following approach to create the key:

1. Pull out all foreign mother companies in Experian and create fake IDs starting with 9999.
2. Pull out all foreign mother companies in Bisnode. Here we already have fake IDs
3. Match these companies on name and country code using step 1,2 and 4 from the matching process of chapter B.5
4. Save the fake IDs from Experian and Bisnode along with the first and last year where the company is in both datasets.

In total, we have managed to find 1,584 foreign companies, that in are linked to 2,129 Danish companies and 5,566 Swedish companies. They have an average lifespan of 5.39 years when they are in both datasets.

F Descriptive Statistics on Cleaned Datasets

Above we described how we constructed a firm-to-firm ownership dataset of all Danish and Swedish firms by merging and linking various datasets. In the first part, we used data from

the Swedish data provider Bisnode on the group structure of Swedish firms. In the second part, we used data from the Danish provider Experian on the group structure of Danish firms. Below, we present some descriptive statistics on the cleaned datasets.

The final Bisnode data includes around over 450 thousand firms (unique ‘od12’), 146 thousand (unique Group mothers, ‘ok12’) and over 3.3 million observations. Table F1 shows some yearly descriptive statistics on the entire Group Register from Bisnode. From the table we can see that in 2018 alone we have 2903 firms and 1104 groups that are Danish.

Table F1: Bisnode: The number of firms/groups (regardless of partner country). All numbers are counts.

	Firms	SE firms	For. firms	Groups	SE Groups	For. groups	DK Firms	DK groups
1995	73345	64381	8964	19733	18145	1588	918	316
1996	85771	74769	11002	24095	22214	1881	1135	427
1997	86701	73668	13033	23944	21901	2043	1280	457
1998	90408	75376	15032	24414	22191	2223	1431	501
1999	93809	77320	16489	25119	22726	2393	1551	550
2000	101380	83272	18108	27298	24483	2815	1798	654
2001	105649	87053	18596	28891	25941	2950	1917	738
2002	109423	89691	19732	30173	26992	3181	2042	794
2003	112516	91735	20781	31179	27977	3202	2165	831
2004	115133	93955	21178	32195	28843	3352	2283	878
2005	122124	99814	22310	34585	31122	3463	2376	953
2006	130535	107231	23304	37172	33414	3758	2516	1025
2007	135779	111023	24756	37934	33907	4027	2633	1095
2008	140165	114521	25644	38870	34704	4166	2713	1158
2009	142778	116880	25898	39802	35454	4348	2661	1153
2010	149106	122794	26312	41762	37405	4357	2637	1124
2011	159208	132262	26946	45054	40554	4500	2708	1164
2012	166965	139575	27390	47348	42673	4675	2762	1168
2013	173841	145886	27955	49291	44531	4760	2791	1166
2014	183090	154360	28730	51649	46748	4901	2842	1175
2015	194252	164674	29578	54463	49403	5060	2903	1186
2016	202047	172444	29603	56131	51232	4899	2872	1138
2017	208705	178620	30085	57274	52354	4920	2864	1108
2018	213277	182061	31216	57485	52471	5014	2903	1114

The final Experian dataset is created by combining the long format dataset with the Swedish ID key and the third country key, which adds the ID numbers found in Bisnode.

Table F2 shows descriptive statistics on the final Experian dataset. After cleaning and reformatting the data, the final Experian data includes around over 290 thousand firms, 116 thousand unique group mothers and over 2.8 million observations. Table F2 shows some yearly descriptive statistics on the entire Group Register from Experian. From the table we can see that in 2018 alone we have 2764 firms and 1533 groups that are Swedish.²⁷

Table F2: Experian: The number of firms/groups (regardless of partner country). All numbers are counts.

	Firms	DK firms	For. firms	Groups	DK Groups	For. groups	SE Firms	SE groups
1995	20190	16091	4099	7460	6168	1292	642	345
1996	23848	18986	4862	8652	7170	1482	762	398
1997	29303	23474	5829	10617	8894	1723	912	468
1998	35893	28848	7045	13089	10918	2171	1100	555
1999	42907	34836	8071	15824	13354	2470	1283	643
2000	49463	40588	8875	18548	15777	2771	1444	724
2001	56707	46196	10511	21301	18099	3202	1654	789
2002	59038	48535	10503	22328	19281	3047	1662	769
2003	65452	53806	11646	25062	21599	3463	1811	809
2004	72209	59339	12870	27813	24020	3793	1936	828
2005	76055	63300	12755	29583	25867	3716	1887	805
2006	83908	70452	13456	32930	28913	4017	1969	810
2007	93960	79697	14263	36978	32766	4212	2049	824
2008	103285	88130	15155	40727	36365	4362	2103	835
2009	111009	94870	16139	43785	39048	4737	2163	859
2010	117204	100296	16908	46352	41253	5099	2248	893
2011	122573	104456	18117	48369	43002	5367	2344	917
2012	125465	106870	18595	49284	43800	5484	2379	954
2013	129331	109723	19608	50823	44906	5917	2488	1032
2014	135317	114904	20413	53250	47160	6090	2577	1111
2015	144848	123558	21290	57267	50650	6617	2670	1229
2016	153930	133282	20648	61407	54732	6675	2658	1352
2017	163840	143177	20663	64837	58483	6354	2693	1423
2018	172911	152093	20818	68219	61766	6453	2764	1533

²⁷See file `Experian_final.ipynb`.